

Data Modeling

Name: Sumit Chauhan

Dept: Data Engineering

- Adding necessary tables from dataset

The screenshot displays the Microsoft Power Query Editor window titled "lms_star_schema". The interface includes a ribbon with tabs: File, Home, Transform, Add Column, View, Tools, and Help. The "Transform" tab is active, showing various data manipulation options like "Choose Columns", "Remove Columns", "Keep Rows", "Remove Rows", "Sort", "Split Column", "Group By", "Data Type: Text", "Use First Row as Headers", "Merge Queries", "Append Queries", and "Combine Files".

On the left, the "Queries" pane shows a list of queries: "Transform File from Sales Data [2]", "Helper Queries [3]" (including "Sample File", "Parameter1 (Sample File)", and "Transform File"), "Other Queries [7]" (including "Sales Data", "Products", "Customers", "Product Categories", "Product Subcategories", "Territories", and "Calendar"). The "Customers" query is currently selected.

The main area displays a data table with the following columns: **CustomerKey**, **Prefix**, **FirstName**, **LastName**, **BirthDate**, and **MaritalStatus**. The table contains 24 rows of data, starting with:

	CustomerKey	Prefix	FirstName	LastName	BirthDate	MaritalStatus
1	11000	MR.	JON	YANG	08-04-1966	M
2	11001	MR.	EUGENE	HUANG	14-05-1965	S
3	11002	MR.	RUBEN	TORRES	12-08-1965	M
4	11003	MS.	CHRISTY	ZHU	15-02-1968	S
5	11004	MRS.	ELIZABETH	JOHNSON	08-08-1968	S
6	11005	MR.	JULIO	RUIZ	05-08-1965	S
7	11007	MR.	MARCO	MEHTA	09-05-1964	M
8	11008	MRS.	ROBIN	VERHOFF	07-07-1964	S
9	11009	MR.	SHANNON	CARLSON	01-04-1964	S
10	11010	MS.	JACQUELYN	SUAREZ	06-02-1964	S
11	11011	MR.	CURTIS	LU	04-11-1963	M
12	11012	MRS.	LAUREN	WALKER	18-01-1968	M
13	11013	MR.	IAN	JENKINS	06-08-1968	M
14	11014	MRS.	SYDNEY	BENNETT	09-05-1968	S
15	11015	MS.	CHLOE	YOUNG	27-02-1979	S
16	11016	MR.	WYATT	HILL	28-04-1979	M
17	11017	MRS.	SHANNON	WANG	26-06-1944	S
18	11018	MR.	CLARENCE	RAI	09-10-1944	S
19	11019	MR.	LUKE	LAL	07-03-1978	S
20	11020	MR.	JORDAN	KING	20-09-1978	S
21	11021	MS.	DESTINY	WILSON	03-09-1978	S
22	11022	MR.	ETHAN	ZHANG	12-10-1978	M
23	11023	MR.	SETH	EDWARDS	11-10-1978	M
24	11024	MR.	BLISS	VIE	17-10-1978	M

The bottom status bar indicates "13 COLUMNS, 999+ ROWS" and "Column profiling based on top 1000 rows". The right-hand "Query Settings" pane shows the "APPLIED STEPS" list, which includes "Source", "Promoted Headers", "Changed Type", and "Removed Duplicates" (the last step is currently selected).

Customers to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

CustomerKey	OrderDate	OrderLineItem	OrderNumber	OrderQuantity	ProductKey	Sales_year
13622	2021-01-03	1	SO48819	1	352	2021
13960	2021-01-13	1	SO48901	1	352	2021
13953	2021-01-14	1	SO48909	1	352	2021

To table

Customers

AnnualIncome	BirthDate	CustomerKey	EducationLevel	EmailAddress	FirstName	Gender
40000	22 April 1973	11479	Bachelors	darryl6@adve...	DARRYL	M
40000	22 May 1972	11483	Bachelors	calvin13@adv...	CALVIN	M
40000	22 November...	11584	Bachelors	devin33@adv...	DEVIN	M

Cardinality Many to one (*:1)

Cross-filter direction Single

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

Save Cancel

Products to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

CustomerKey	OrderDate	OrderLineItem	OrderNumber	OrderQuantity	ProductKey	Sales_year
13622	2021-01-03	1	SO48819	1	352	2021
13960	2021-01-13	1	SO48901	1	352	2021
13953	2021-01-14	1	SO48909	1	352	2021

To table

Products

ModelName	ProductColor	ProductCost	ProductDescri...	ProductKey	ProductName	ProductPr
Sport-100	Red	13.0863	Universal fit, ...	214	Sport-100 Hel...	34.99
Sport-100	Black	12.0278	Universal fit, ...	215	Sport-100 Hel...	33.6442
Sport-100	Blue	12.0278	Universal fit, ...	220	Sport-100 Hel...	33.6442

Cardinality Many to one (*:1)

Cross-filter direction Single

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

Save Cancel

Territories to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

OrderItem	OrderNumber	OrderQuantity	ProductKey	Sales_year	StockDate	TerritoryKey
	SO48819	1	352	2021	2020-09-06	9
	SO48901	1	352	2021	2020-09-19	9
	SO48909	1	352	2021	2020-09-22	9

To table

Territories

Continent	Country	Region	SalesTerritor...
North America	United States	Northwest	1
North America	United States	Northeast	2
North America	United States	Central	3

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

Calendar to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

CustomerKey	OrderDate	OrderLineItem	OrderNumber	OrderQuantity	ProductKey	Sales_year
13622	03 January 20...	1	SO48819	1	352	2021
13960	13 January 20...	1	SO48901	1	352	2021
13953	14 January 20...	1	SO48909	1	352	2021

To table

Calendar

Date	MonthName	MonthNumber	StartOfMonth	StartOfWeek	StartOfYear	Year
05 January 20...	January	1	1	2	01 January 20...	2020
06 January 20...	January	1	1	2	01 January 20...	2020
07 January 20...	January	1	1	2	01 January 20...	2020

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

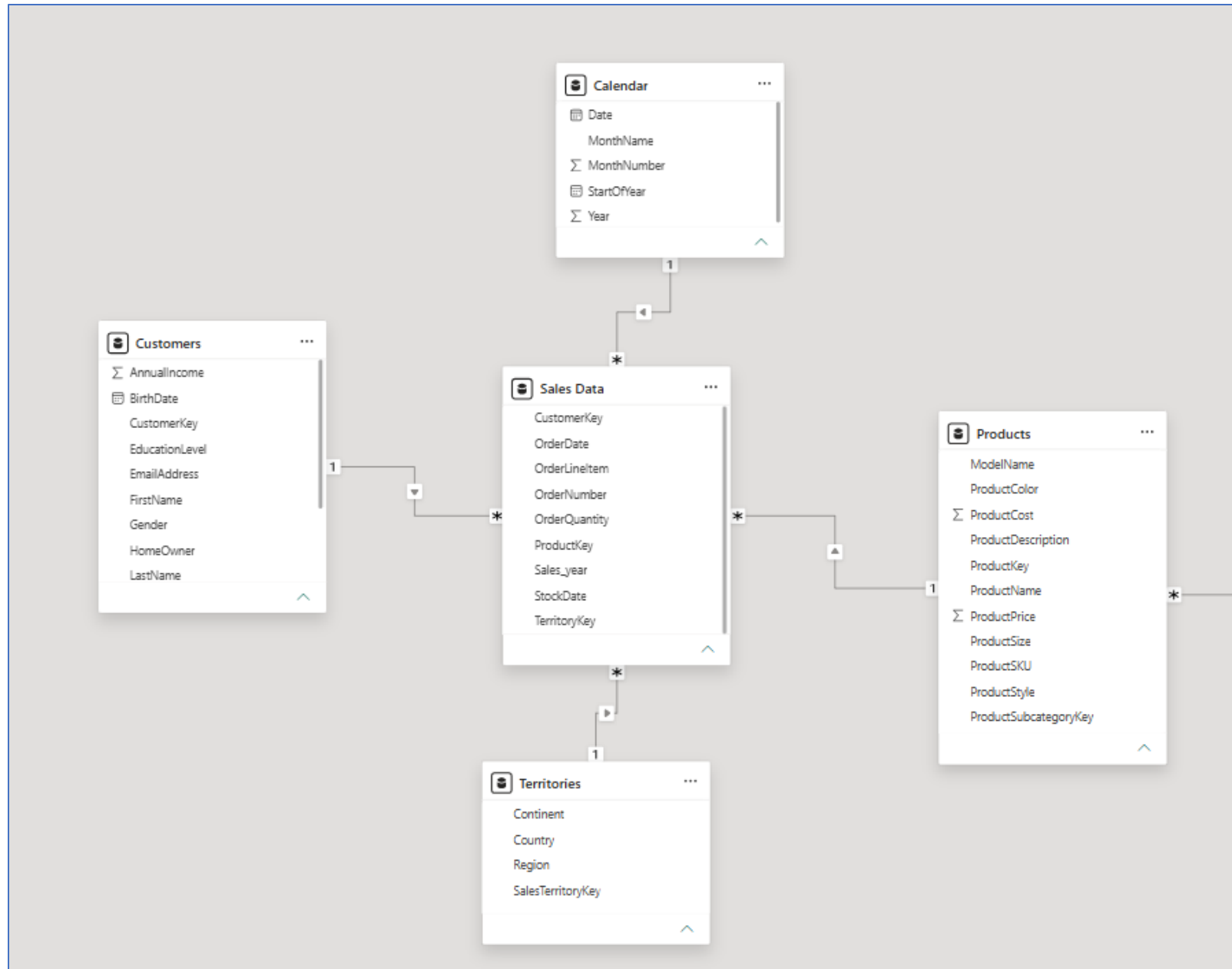
☐ Apply security filter in both directions

☐ Assume referential integrity

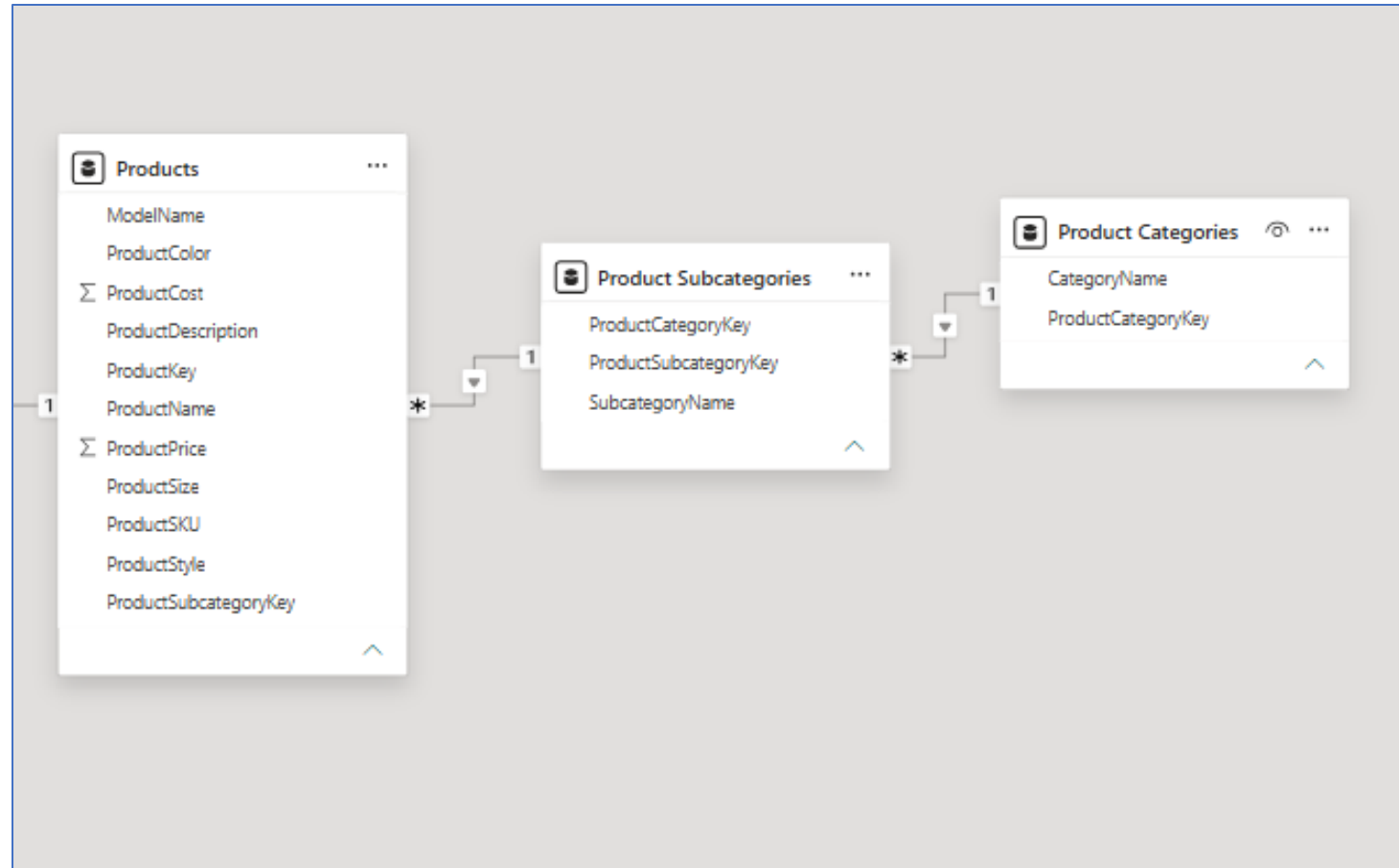
Save

Cancel

Star Schema



Products – subcategories – categories



Adding necessary columns in Calender table for hierarchy

Custom Column

Add a column that is computed from the other columns.

New column name

StartOfMonth

Custom column formula ⓘ

= DateDate.StartOfMonth([Date])

Available columns

Date

MonthName

MonthNumber

StartOfYear

Year

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

StartOfWeek

Custom column formula ⓘ

= Date.StartOfWeek([Date], Day.Monday)

Available columns

Date

MonthName

MonthNumber

StartOfYear

Year

StartOfMonth

<< Insert

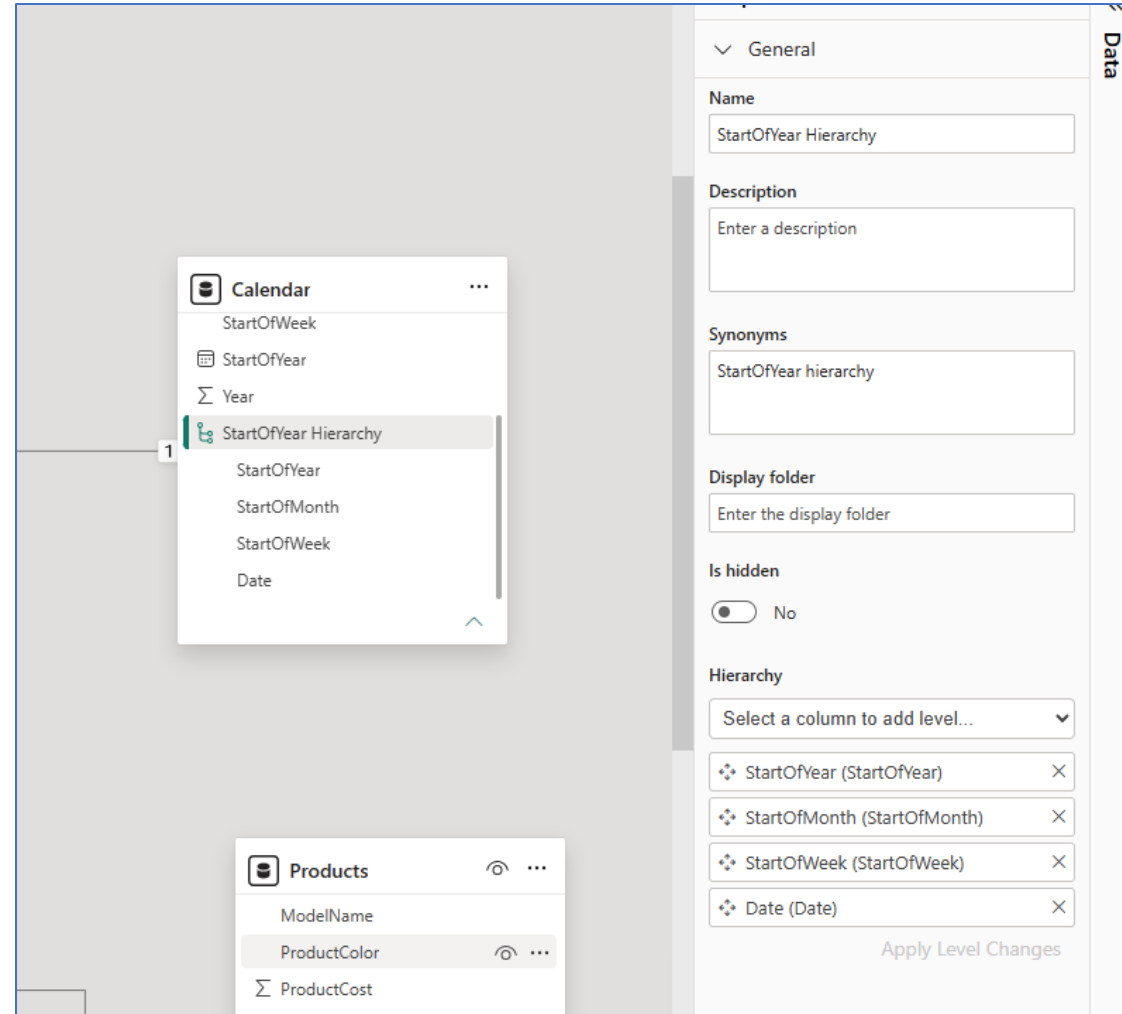
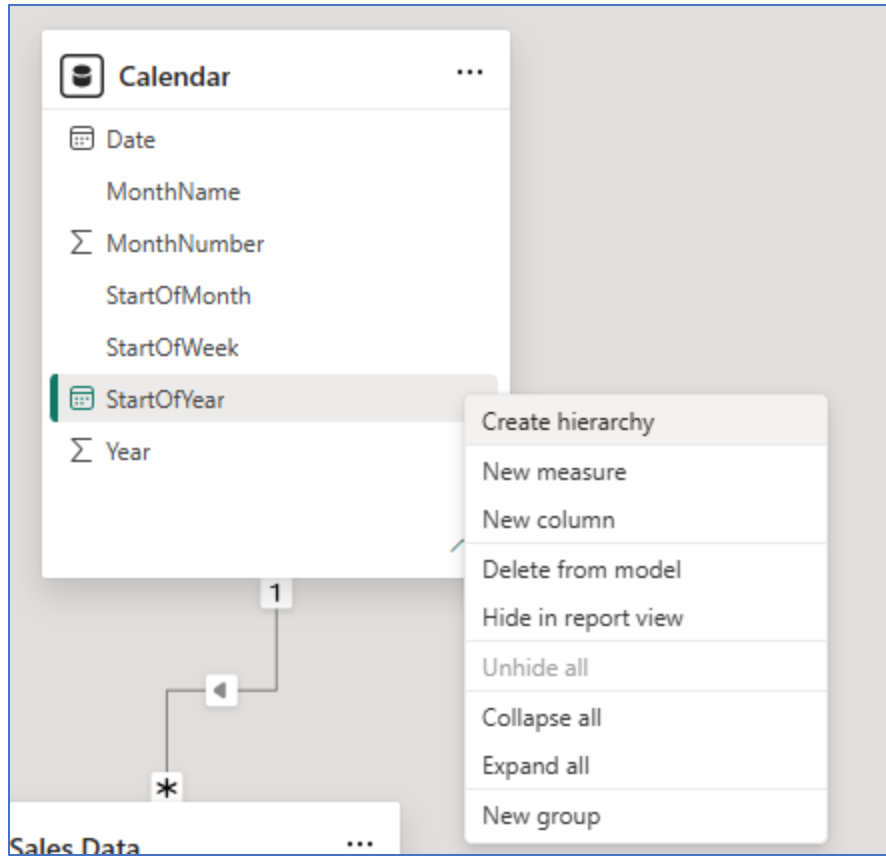
[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Creating hierarchy from startOfYear



Matrix visual for drill up/down

The screenshot displays the Microsoft Power BI Desktop interface. The ribbon at the top includes tabs for File, Home, Insert, Modeling, View, Optimize, Help, Format, and Data / Drill. The Data / Drill tab is active, showing options for New visual, Text box, More visuals, New visual calculation, New measure, Quick measure, Sensitivity, Publish, and Prep data for Copilot AI.

The main area shows a matrix visual with the following data:

StartOfWeek	Black	Blue	Multi	NA	Red	Silver	White	Yellow	Total
1	1366	524	811	7100	552	437	140	665	11595
2	2485	863	1416	11965	871	797	224	1080	19701
3	2421	877	1240	11582	973	767	280	1114	19254
4	2461	888	1329	11726	908	723	229	1077	19341
5	1765	599	897	8279	674	508	175	667	13564
6	92	28	63	428	33	25	15	35	719
Total	10590	3779	5756	51080	4011	3257	1063	4638	84174

The right-hand pane shows the Filters, Visualizations, and Data sections. The Filters section includes filters for Date, ProductColor, StartOfMonth, StartOfWeek, Sum of OrderQuantity, Year, and Add data fields here. The Visualizations section shows the matrix visual selected, with Rows set to Date Hierarchy (Year, StartOfMonth, StartOfWeek, Date) and Columns set to ProductColor. The Values section shows Sum of OrderQuantity. The Data section shows a search bar and a list of data sources: Calendar, Customers, Product Categories, Product Subcategories, Products, Sales Data, and Territories.

Drill up/down on metrix visual with date hierarchy as rows and sum of productQuality as values

Date	Black	Blue	Multi	NA	Red	Silver	White	Yellow	Total
01 January 2020	3				1				4
02 January 2020					4				4
03 January 2020	2				5	1			8
04 January 2020					4	1			5
05 January 2020					3				3
06 January 2020	1				5				6
07 January 2020					4				4
08 January 2020	2				5	1			8
09 January 2020					4				4
10 January 2020	1				3				4
11 January 2020	2				6	1			9
12 January 2020					7				7
13 January 2020	1				2				3
14 January 2020					6	3			9
15 January 2020	3				3	1			7
Total	10590	3779	5756	51080	4011	3257	1063	4638	84174

StartOfMonth	Black	Blue	Multi	NA	Red	Silver	White	Yellow	Total
1	872	316	506	4587	409	276	105	375	7446
10	800	277	437	4256	224	225	103	288	6610
11	761	299	421	4152	257	212	91	330	6523
12	1063	380	577	5006	306	307	101	509	8249
2	888	322	529	4366	391	285	85	394	7260
3	943	371	570	4643	425	328	94	417	7791
4	1027	384	536	4909	446	301	91	480	8174
5	1132	422	552	5109	469	398	96	556	8734
6	1131	421	639	5041	443	421	109	579	8784
7	429	126	134	1036	159	142	11	164	2201
8	765	229	432	4048	256	166	89	251	6236
9	779	232	423	3927	226	196	88	295	6166
Total	10590	3779	5756	51080	4011	3257	1063	4638	84174

StartOfWeek	Black	Blue	Multi	NA	Red	Silver	White	Yellow	Total
1	1366	524	811	7100	552	437	140	665	11595
2	2485	863	1416	11965	871	797	224	1080	19701
3	2421	877	1240	11582	973	767	280	1114	19254
4	2461	888	1329	11726	908	723	229	1077	19341
5	1765	599	897	8279	674	508	175	667	13564
6	92	28	63	428	33	25	15	35	719
Total	10590	3779	5756	51080	4011	3257	1063	4638	84174

Year	Black	Blue	Multi	NA	Red	Silver	White	Yellow	Total
2022	5062	2236	3332	28655	1281	1573	580	2595	45314
2021	4676	1543	2424	22425	1374	1406	483	1899	36230
2020	852				1356	278		144	2630
Total	10590	3779	5756	51080	4011	3257	1063	4638	84174